TD PI CONNECTOR

User Manual

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Introduction

The TD PI Connector is a Windows Service that syncs data from the PI System to TDEngine, which is an open source, cloud-native time series database (TSDB) optimized for Internet of Things (IoT).

After setting up TD PI Connector configuring files, when the Windows Service starts, it creates all tables on TDEngine and sends data from the PI Data Archive to TDEngine. This connector is compatible with TDEngine hosted on-premise and in the cloud.

The connector once it starts, it retrieves the latest row from each table. This enables the connector to backfill past data in case it is needed. This is a great feature added to prevent data loss.

There are some applications that are installed together with the TD PI Connector:

- CSV Point Builder: a Windows Form application that makes it easy for the user to select the PI Point list to send data to TDEngine
- TDBackfill: a Console Application to backfill old data to TDEngine.

The TD PI Connector works using PI Point Mode or AF Element Mode. PI Point mode creates one TDEngine table per PI Point/AF Attribute and the AF Element mode creates a table per AF Element.

The following sections of this user manual provide more details about the product.

Installation

Prerequisites

The prerequisites to install TD PI Connector are below:

- .NET Framework 4.7.2+
- PI AF Client 2018+
- TD Engine 3.0.1.5+

Setup Wizard

Download TDEngine.Setup.msi and run it with admin privileges. The TD PI Connector Wizard will appear. Click on "Next".

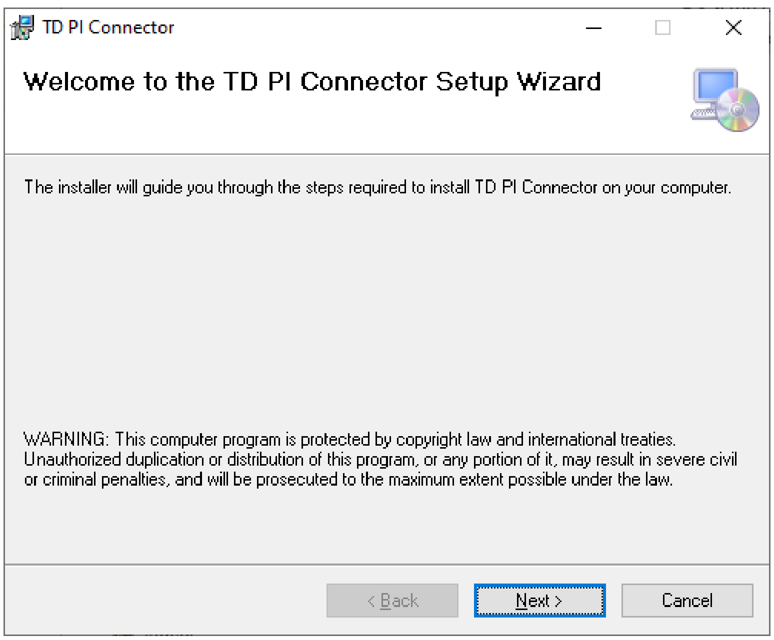

Figure 1 – TD PI Connector Setup Wizard

Select the folder which the application will be installed. By default, the folder is "C:\Program Files (x86)\TD PI Connector". Click on "Next". Then click on "Next" again to start copying files and "Close" to close the setup wizard.

Open the Service and open the TD PI Connector Properties as shown on Figure 2.

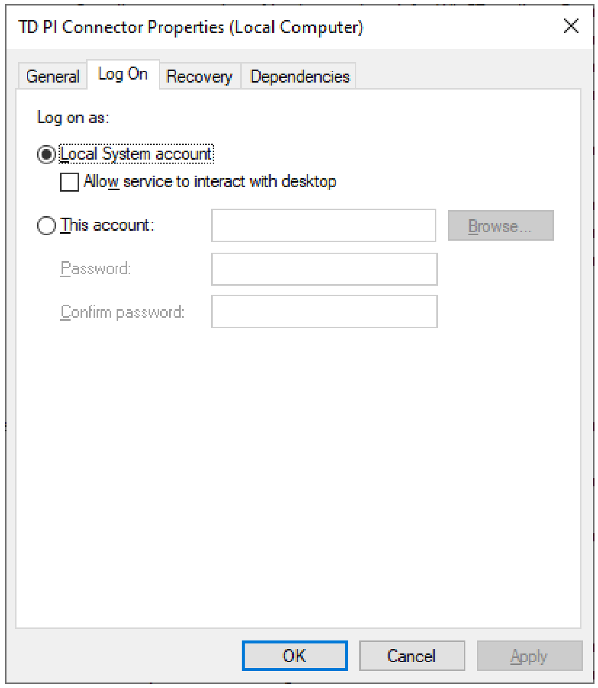

Figure 2 – TD PI Connector Properties

The service can run using local system account or a custom domain account as long as the account has enough privileges to connect and write data to PI Data Archive, PI AF and the TDPIConnector folder.

Do not start the service until the configuration files are configured. Please refer to the next section for more information.

Setting up the configuration files

TDPIConnector.Service.exe.config

Once TD PI Connector installed, some configure files need to be edited.

Open TD PI Connector\TDPIConnector.Service.exe.config file and update the parameter values according to the description below:

- UpdateInterval – Number in milliseconds for each cycle for getting updated values. Default: 10000.
- PISystemName – AF Server hostname
- PIServerName – PI Data Archive hostname
- AFDatabaseName – AF Database name
- TDEngineHost – TDEngine host
- TDEnginePort – TDEngine port number
- TDEngineToken – TDEngine token (cloud only)
- TDEngineUsername – TDEngine username (on premise only)
- TDEnginePassword – TDEngine password (on premise only)
- TDEnginePIDatabase – TDEngine database name to sync PI data. Default: pi.
- WebBaseUrl – Host of the monitoring & troubleshooting portal. Default: http://localhost.
- WebBasePort – Port number of the monitoring & troubleshooting portal. Default: 9000.
- MaxPIEvents – Maximum number of events stored by the monitoring & troubleshooting portal to display to the user.
- MaxTDEngineHttpResponses – Maximum number of HTTP responses stored by the monitoring & troubleshooting portal to display to the user.

The last four parameters are related to monitoring & troubleshooting portal which is described later in this user manual.

log4net.config

If you want to change the log debug level of the application, open log4net.config:

```
<log4net>
  <appender name="ColoredConsoleAppender"
type="log4net.Appender.ColoredConsoleAppender">
    <mapping>
      <level value="ALL" />
 ...
```

You might want to change ALL to ERROR in case you are interested in seeing only errors and not informational messages.

PI Point and AF Element modes

The TD PI Connector works using PI Point Mode or AF Element Mode. PI Point mode creates one TDEngine table per PI Point/AF Attribute and the AF Element mode creates a table per AF Element.

PI Point Mode

The Points.csv and ElementTemplate1.csv files are configured for using the PI Point mode. The workflow of each file is below:

Points.csv (working with PI Points directly)

- When TD PI Connector starts, it loads the PI Point names from the Points.csv file.
- It creates all the TDEngine tables on the TDEngine database defined on the TDEnginePIDatabase parameter using the PI Point list, one table per PI Point.
- It starts to send data from the PI Data Archive to TDEngine.
- The CSV Point Builder is a great tool to quickly generate the Point.csv file.

ElementTemplate1.csv (working with AF assets)

- When TD PI Connector starts, it loads the AF Element Template names from the ElementTemplate1.csv file.
- It searches for all AF element instances derived from each AF Element Template.
- For each AF element found, it gets all their AF Attributes using the PI Point Data Reference.
- It creates a list of PI Points that the AF Attribute are mapping to.
- It creates all the TDEngine tables on the TDEngine database defined on the TDEnginePIDatabase parameter using the PI Point list, one table per PI Point/AF Attribute.
- It starts to send data from the PI Data Archive to TDEngine.

Both types of CSV files create a TDEngine table per PI Point with following columns:

- ts (DATETIME),
- val(NCHAR/INT/FLOAT/DOUBLE/TIMESTAMP)
- quality(INT)

The type of the val column created by the TD PI Connector depends on the PI Point Type. Table 1 shows how the TD PI Connector chooses the TDEngine column type of the val column according to point type.

Point Type	TDEngine Column Type
Digital	NCHAR
Int16	SMALLINT
Int32	INT
Int64	BIGINT
Float16	FLOAT
Float32	FLOAT
Float64	DOUBLE
String	NCHAR
Timestamp	TIMESTAMP

Table 1 – Mapping between the selected Point Type and the TDEngine Column Type

AF Element Mode

The ElementTemplate2.csv file is configured for using the AF Element mode. According to TDEngine, this mode should be used when all the AF Attributes of each AF Element receive data with same timestamps. The workflow is below:

ElementTemplate2.csv (working with AF assets)

- When TD PI Connector starts, it loads the AF Element Template names from the ElementTemplate2.csv file.
- It searches for all AF element instances derived from each AF Element Template.
- It creates all the TDEngine tables on the TDEngine database defined on the TDEnginePIDatabase parameter using the AF Element list, one table per AF Element.
- It starts to send data from the PI Data Archive to TDEngine.

TD PI Connector creates a TDEngine table per AF Element with following columns:

- ts (DATETIME),
- {attribute1_name}_val(NCHAR/INT/FLOAT/DOUBLE/TIMESTAMP)
- {attribute1_name}_status(INT)
- {attribute2_name}_val(NCHAR/INT/FLOAT/DOUBLE/TIMESTAMP)
- {attribute2_name}_status(INT)
- The other attribute columns

Let's suppose there is an AF Element name "Chicago" and it has two attributes "Pressure" and "Temperature". The TDEngine table for this mode would have the following columns:

- ts (DATETIME),
- pressure_val(FLOAT)
- pressure_status(INT)
- temperature_val(FLOAT)
- temperature_status(INT)

Note that the column type of the TDEngine columns ending with "_val" are also selected according to the table 1.

CSV Point Builder

CSV Point Builder is a Windows Form application that helps the user quickly create/edit the Points.csv file by searching for PI Points on the PI Data Archive.

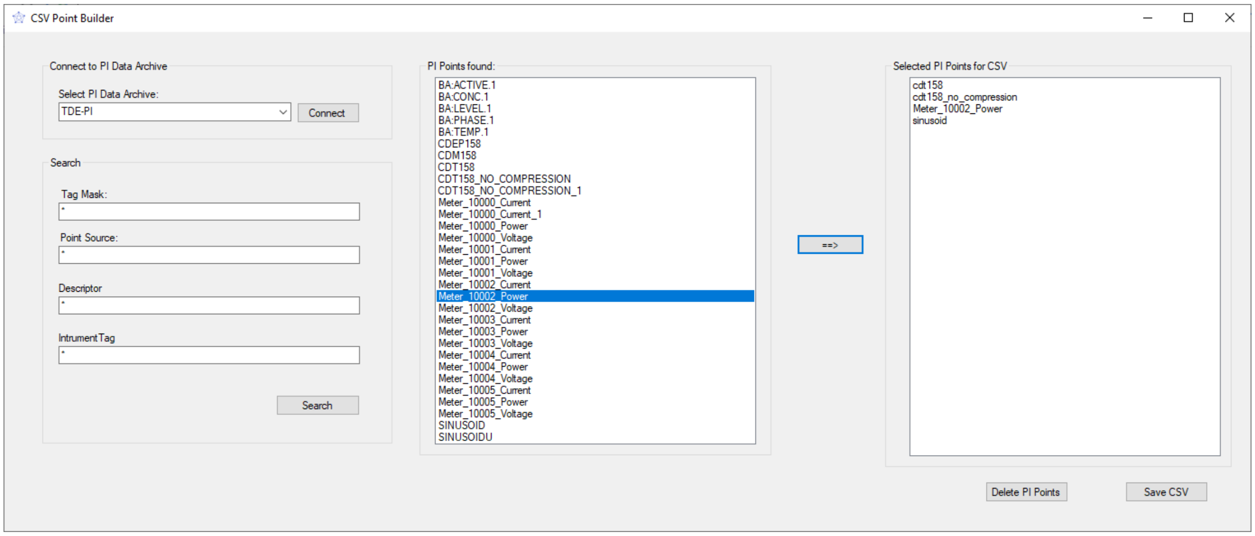

Figure 3 – CSV Point Builder

Steps to use the CSV Point Builder:

1. When the application is loaded, the PI Points (stored on the Points.csv file) are shown on the right list box.
2. Select the PI Data Archive and click on the “Connect” button.
3. Edit the tag mask, point source, descriptor and instrument tag fields to search for PI Points. Clicks on the “Search” button.
4. The list of PI Points found in the search are shown on the left list box of the application.
5. Select each PI Point (holding the CTRL button) to add to the CSV file and click on the “== >” button.
6. Click on the “Delete PI Points” to delete the selected PI Points (holding the CTRL button) of the right list box.
7. Click on the “Save CSV” button to save the CSV file with the selected PI Points.

Backfill

Backfill with TD Backfill

If the user wants to backfill past data to TDEngine before sending real-time values, TDBackfill is the tool to achieve this goal. The console application is also installed with TD PI Connector.

Before starting to use this tool, the TD PI Connector\TDBackfill.exe.config file needs to be edited using the same key/values of the TDPIConnector.Service.exe.config, described on page 5.

Open the command prompt and type: "TDBackfill –help" to visualize general instructions about how to use this tool as shown on Figure 4.

```
C:\Windows\system32\cmd.exe - TDBackfill  --help
C:\Program Files (x86)\TD PI Connector>TDBackfill --help
2022-10-27 09:40:22,509 [1] INFO  TDBackfill.Program - test
Help Message:
Usage: TDBackfill.exe [options]
Options:
-h, --help
    Display this help message.
-drop, --drop-table
    Drop the associated table before backfilling.
    This will delete all data in the table. Ignored if -t or -f are specified.
-a, --all
    Backfill all data.
-t, --to-first-recorded
    Backfill up to the first recorded value in TDengine.
-f, --from-last-recorded
    Backfill from the last recorded value in TDengine.
-s, --start
    The start time for backfilling data. If not provided, the start time will be the earliest time available.
-e, --end
    The end time for backfilling data. If not provided, the end time will be the current time.
Examples:
    TDBackfill.exe -drop -a
    TDBackfill.exe -t
    TDBackfill.exe -f
    TDBackfill.exe -drop -s 1/1/2014 -e 1/2/2014
    TDBackfill.exe -t -s "1/1/2014 10 am"
    TDBackfill.exe -f -e "1/2/2014 14:32:12"
2022-10-27 09:40:22,522 [1] INFO  TDBackfill.Program - test
Press any key to exit...
```

Figure 4 – TDBackfill console application help instructions

Backfill with TD PI Connector

The connector once it starts, it retrieves the last row from each table. If the connector detects a data loss within an interval, it will backfill past data. This is a great feature added to prevent data loss. Note that the TD PI Connector backfill data having the most recent value in TDEngine table as the starting time of the backfill. If the user wants to migrate older data, TDBackfill is the suitable tool for this.

Monitoring && Troubleshooting

The TD PI Connector Service exposes a web portal for monitoring && troubleshooting on the port number specified on the WebBasePort parameter. The portal has 3 main pages:

- Monitoring
- Logs
- Exceptions

Monitoring

This page provides general information about the items below as shown on Figure 5:

- PI Data Archive (Host, port, server version)
- TD Engine (Host, port, server version)
- PI Data Archive Events → List of events with PI data received by PI Data Archive. If the application is not receiving values for a long time, it probably means that something is not working properly.
- TD Engine HTTP status code responses → List of HTTP status code responses of the HTTP requests made by the PI Data Archive against TDEngine. If the HTTP status code is not 201, then something is not working properly.

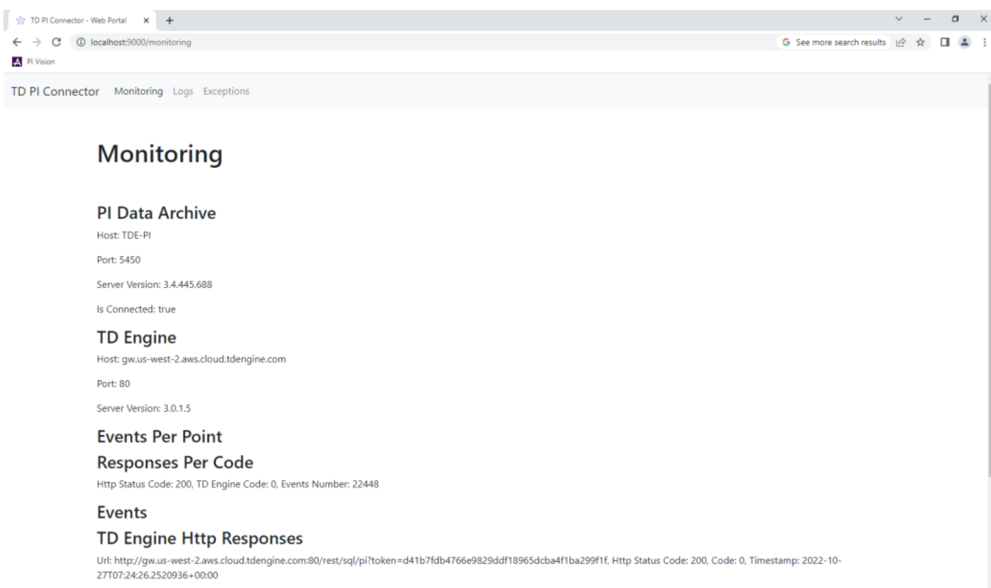

Figure 5 – Monitoring page of the web portal

Logs

This page allows the users to open the logs of the system (as shown on Figure 6) without having to go to \TDPIConnector\Logs folder.

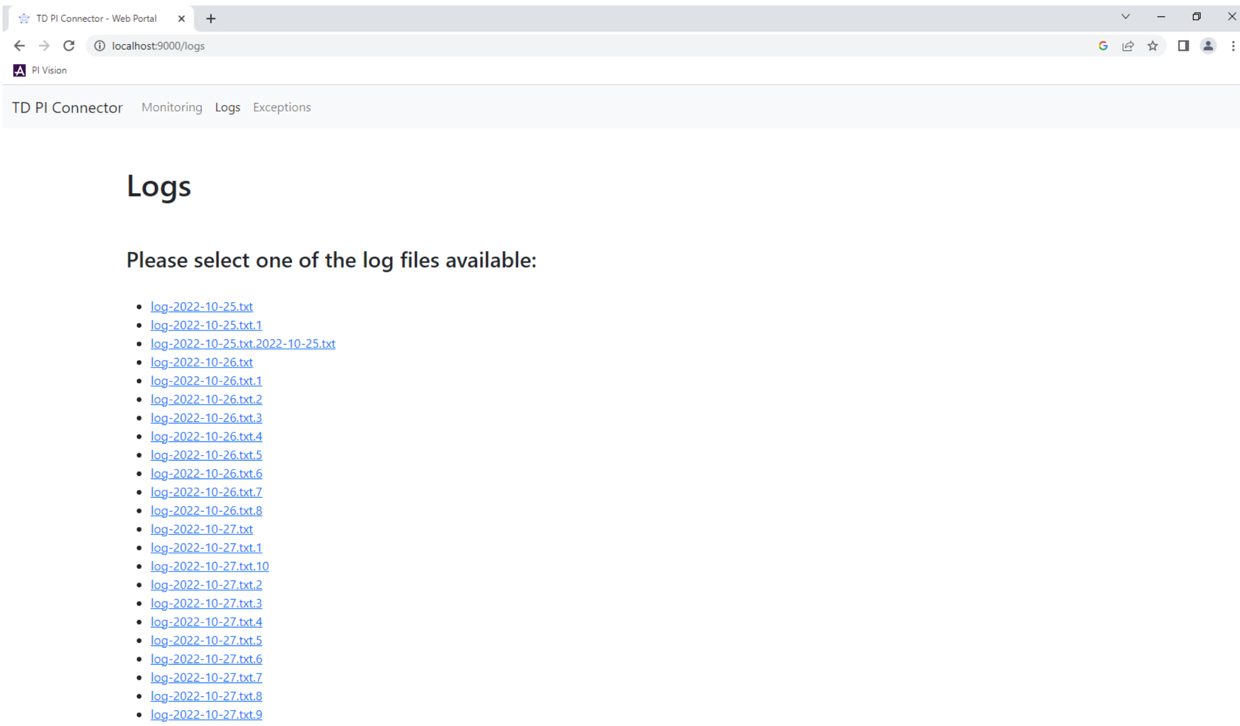

Figure 6 – Logs page of the web portal

Exceptions

The exception page makes it easier for the users to visualize all the exceptions thrown in the application (since the Windows Service has been started) without having to open the logs of the system. The logs contain a lot of information (not only exceptions) so having a page that filters the exceptions could save a lot of the user's time.

TDengine Support

If you are an Enterprise Edition customer or interested in enterprise services, contact business@tdengine.com to speak with an account representative.

For more information about TDengine, you can follow us on social media and join our Discord server.